Alexandre Minets

Nationality: French, Belarusian
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Research interests

Geometric representation theory, moduli spaces of sheaves, cohomological Hall algebras, KLR algebras, quantum groups, Higgs bundles, quiver varieties.

Employment

2026-present Postdoctoral fellow at the *University of Bonn*, Germany
2023-2025 Postdoctoral fellow at *Max Planck Institute*, Bonn, Germany
2021-2023 Postdoctoral research associate at the *University of Edinburgh*, UK
2018-2021 Postdoctoral researcher at Hausel Group, *IST Austria*, Klosterneuburg, Austria

Education

2015-2018 PhD student in Mathematics, *Université Paris-Sud*, Orsay, France
Topic: "Cohomological Hall algebras and Nakajima varieties associated to curves"
defended on *September 3, 2018* | advisor: Olivier Schiffmann
2012-2015 Élève at *École Normale Supérieure*, Paris, France
2014 Master 2 in Mathematics, mention "très bien" (GPA 18/20), *Paris 7*
2013 Master 1 in Mathematics, mention "très bien" (GPA 17,8/20), *Paris 6*
2009-2012 Student in Applied Mathematics, *Belarusian State University*, Minsk, Belarus

Papers

- Quantum wreath products and Yang–Baxter equations, *preprint*, arXiv:2609.02765
- Equivariant multiplicities and mirror symmetry for Hilbert schemes (joint with F. Živanović), *submitted*, arXiv:2602.14897
- Stratifying quiver Schur algebras via ersatz parity sheaves (joint with R. Maksimau), *submitted*, arXiv:2504.17430
- Schur-Weyl duality for quantum wreath products (joint with C.-J. Lai), *accepted to Representation Theory*, arXiv:2502.02108
- Coherent sheaves on surfaces, COHAs and deformed $W_{1+\infty}$-algebras (joint with A. Mellit, O. Schiffmann, E. Vasserot), *Moduli, Volume 3, 2026, e13*, DOI: 10.1017/S2949764726100277
- $P = W$ via $\mathcal{H}_2$ (joint with T. Hausel, A. Mellit, O. Schiffmann), *accepted to JAMS*, arXiv:2209.05429
- KLR and Schur algebras of curves and semi-cuspidal representations (joint with R. Maksimau), *International Mathematics Research Notices, Volume 2023, Issue 8, April 2023, Pages 6976-7052*, DOI: 10.1093/imrn/rnac055
- Cohomological Hall algebras for Higgs torsion sheaves, moduli of triples and sheaves on surfaces, *Selecta Mathematica (N.S.) 26, No. 2, Paper No. 30, 67 p. (2020)*, DOI: 10.1007/s00029-020-00553-x

Funding and distinctions

- Séminaire Bourbaki on proofs of $P = W$ conjecture (2023)
- LMS Research in Pairs (2023): £900 for collaboration with R. Maksimau (University of Cergy-Pontoise)
- Interviewed for EPSRC Postdoctoral Fellowship (2023; ranked 2nd, 1 application funded)

Invited talks: conferences and workshops

2026 HKUST, conference "Enumerative Geometry in East Asia 2026" (forthcoming)
MFO, workshop "Character Theory and Categorification"
University of Konstanz, DMV Annual Meeting
ICMS Sofia, conference "Quantum Groups and Cluster Algebras II"
SIMIS, conference "Quantum Algebras Meet Gauge Theory and String Theory"
2025 TU Munich, ARTIG 7
2024 CIRM, Colloque Tournant
2023 OIST, workshop "Representation Theory of Hecke Algebras and Categorification"
2021 MFO, workshop "Enveloping Algebras and Geometric Representation Theory"
2020 Institut Henri Poincaré, Colloque Tournant
2019 USTC Hefei, Séminaire MathJeunes
University of Edinburgh, conference "Geometric Representation Theory and Low-Dimensional Topology"
University of Edinburgh, ARTIN 2019
2017 Isle of Wight, British Isles Graduate Workshop (expository)
RIMS, Algebraic Analysis and Representation Theory Conference (poster)

Invited talks: seminars

2026 University of Paderborn, Algebra Seminar
MPIM Bonn, Unterseminar (expository)
Aix-Marseille University, RGR Seminar
2025 HKUST, Algebra and Geometry Seminar
Academia Sinica, Representation Theory Seminar
University of Amiens, Séminaire GAT
University of Bonn, Representation Theory Seminar
2024 University of Cergy-Pontoise, Séminaire de Géométrie
University of Strasbourg, Séminaire ART
University of Bielefeld, BIREP Seminar
University of Mainz, Oberseminar Algebraische Geometrie
University of Frankfurt, Oberseminar Algebra und Geometrie
KSU, M-seminar (online)
2023 MPI Bonn, Oberseminar
Academia Sinica, Algebraic Geometry Seminar (2 talks)
HU Berlin, Arithmetic Geometry Seminar
CUHK, Representation and Number Theory Seminar
HKUST, Algebra and Geometry Seminar
ANU, MSI Colloquium (2 talks)
University of Melbourne, Representation Theory Seminar
IPMU, Mathematics and String Theory Seminar
University of Liverpool, Algebraic Geometry Seminar
2022 EPFL, Algebraic Geometry Seminar
University of Edinburgh, Hodge Seminar
2021 University of Edinburgh, EGRET Seminar
University of Edinburgh, Hodge Seminar
2020 RepNet Virtual Seminar
2019 University of Stuttgart, Algebra Seminar
Academia Sinica, Postdoc Seminar
Universität Bonn, Representation Theory Seminar
Université de Montpellier, AGATA Seminar
2018 IST Austria, Algebraic Geometry Seminar
IST Austria, Working Seminar
2017 Institut Henri Poincaré, Seminaire RéGA (expository)
2015 University of Wuppertal, School on Kac conjectures and quiver varieties (expository)
École Normale Supérieure, Seminaire MathJeunes (expository)

Teaching experience

2024 Graduate course "Equivariant methods in representation theory" (designer, lecturer, 100%)
University of Bonn, Germany

2023 Graduate course "Homological algebra" (co-lecturer, 30%)
Graduate course "Manifolds" (lecturer, 100%)
University of Edinburgh, UK

2020 Graduate course "$\mathcal{D}$-modules" (designed, lecturer, 50%)
IST Austria, Klosterneuburg, Austria

2015-2017 Teaching assistant, *Université Paris-Sud*, Orsay, France:
- Calculus and probability theory (3 times)
- Linear algebra and differential equations (2 times)
- Multivariate calculus and optimization

2011 Special course in algebraic topology (co-lecturer with S. Patotski)
Belarusian State University, Minsk, Belarus

Other activities

- Organized a 4-day research meeting (20 people), University of Bonn (2025)
- Organized Algebra seminar, University of Edinburgh (2023)
- Organized a reading group on stable envelopes, University of Edinburgh (2022)
- Refereed articles for Publ. math. de l'IHÉS, Compositio Math., Forum Math. Sigma and others
- Participated in pre-screening of PhD student applications, IST Austria (2020, 2021)
- Supervised student project "Atiyah's classification of vector bundles on an elliptic curve", IST Austria (2020)
- Organized a reading group "Number fields with given ramifications and automorphic forms", Université Paris-Sud (2017)
- Member of jury at TFJM^2 (2017, 2018) and ITYM (2018)
- Member of selection committee and teaching assistant at MoMISSS summer school, Bremen, Germany (2013, 2017) and Lyon, France (2014)

Languages:

English: fluent Russian: native
French: fluent German: basic